

Fitness Tracker

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Introduction

Fitness trackers are now on many people's wrists, helping them count steps, check heart beats, and track sleep. They've become needed tools in daily life because they help us understand our health better. With a quick look at your wrist, you can see if you have moved enough today or if your heart is working too hard. Trackers remind busy people to stand up and move when they've sat too long. They also help to show patterns over time, such as if you sleep better on days when you exercise. For people who are trying to be healthier, these devices provide the numbers and facts that keep them going. In this project, we will see how to track various gym exercises like bench press, squats, and more.

Machine Learning Role

Machine learning helps users automatically track their gym workouts by recognizing different exercises like bench press or squats through motion data from sensors such as accelerometers and gyroscopes. Instead of manually logging exercises, users get real-time exercise detection. This makes it easier to stay consistent, monitor progress, and stay motivated throughout your fitness journey.

Objective

The objective of this project is to develop a machine learning model capable of recognizing and classifying gym exercises in real time using sensor data from accelerometers and gyroscopes. To achieve this, we will experiment with various models, including Support Vector Machines (SVM), Random Forest, and others, to identify which approach offers the best performance in predicting exercises. One challenge is making sure the model works well for different people, exercise types, and movement styles, so it can be used by anyone.

Dataset Description

The dataset consists of motion sensor data from 5 participants performing a variety of fitness exercises. The data includes accelerometer and gyroscope readings, with each set labeled according to the specific exercise. The data set covers six different exercises, including bench press, overhead press, squat, deadlift, row, and rest, and participants perform varying numbers of sets for each exercise.

The dataset includes the following variables:

- **Epoch (ms):** Millisecond timestamp of each sensor reading.
- **Accelerometer_x/y/z:** Measures body acceleration in three directions.
- **Gyroscope_x/y/z:** Measures rotational velocity in three directions.
- **Participant:** The person performing the exercise.
- **Label:** The specific exercise performed (e.g., squat).
- **Category:** Intensity level (e.g., light, medium, heavy).
- **Set:** Identifier for each repetition set.

Studying the Dataset

In this project, we work with real-time movement data collected every 200 milliseconds (epoch) using wearable devices. These devices include two main sensors: an accelerometer and a gyroscope. The dataset includes data from multiple participants, each performing a variety of exercises from different categories. This variety helps train machine learning models to recognize different movement patterns across different people.

Accelerometers: measure how fast something is speeding up or slowing down in three directions: up/down, left/right, and forward/backward (X, Y, Z axes). This helps capture simple body movements like walking, jumping, or lifting weights. When you move your arm or leg, the accelerometer picks up that motion as changes in speed.

Gyroscopes: measure how something is rotating or turning around the same three directions. They track rotation and balance, such as when you twist your body, bend forward, or turn to the side. This helps detect more complex movements that involve changes in body position.

Together, these sensors give us a full picture of what the body is doing both in terms of straight movement and rotation. By using the signals they produce, we can train a machine learning model to recognize which exercise a person is performing at any moment.

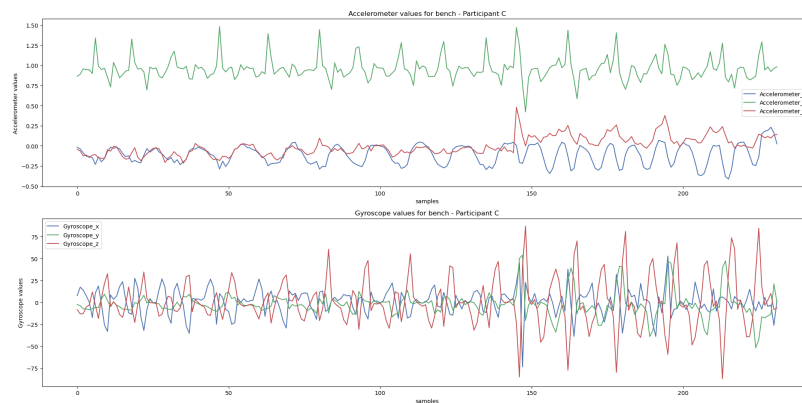


Figure 1: Accelerometer and Gyroscope Data from Participant.

Why This Dataset?

This dataset is a time series, it shows how movements change over time. This is great for tracking exercises. It has detailed data from accelerometers and gyroscopes, so we can see both straight and turning movements. The data is recorded very often, which makes it good for real-time predictions. It also helps learning how to work with signals. For example, trying things like smoothing the data or using sliding windows to get it ready for machine learning.

Data Preprocessing

Before using the sensor data, we need to clean it to make sure it's accurate and useful. One common problem is outliers unusual values caused by sensor errors, random noise, or sudden unexpected movements. By correcting these outliers, we avoid confusing the machine learning models and improve the accuracy.

Outliers

We analyze accelerometer and gyroscope data during exercises. Figures 2 show the boxplots.

Gyroscope: For example in a squat, the body moves mostly up and down, not much turning. So the gyroscope values stay close to zero. Even a small twist shows up as an outlier because it stands out from the rest of the data. This is why we see many outliers the normal values are tightly grouped, so small changes look big.

Accelerometer: Squats involve big up-and-down movements, which the accelerometer captures. The median line may not be in the center, showing that the motion isn't perfectly balanced. But since large changes are expected, we see fewer outliers.

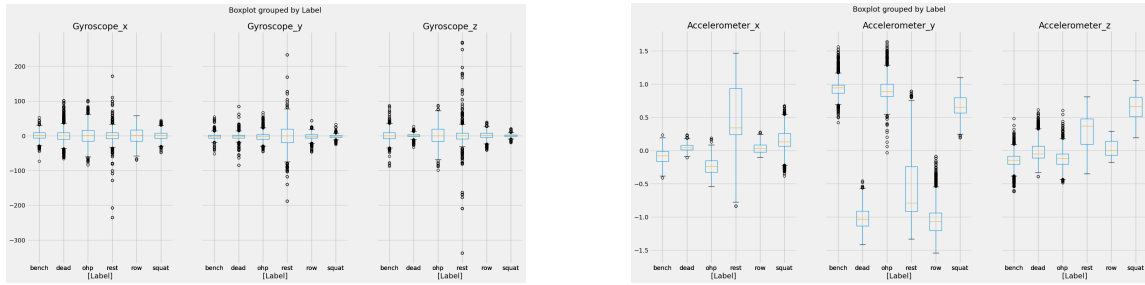


Figure 2: Outliers detected Gyroscope, Accelerometer.

Detecting Outliers with DBSCAN

To detect outliers, using **DBSCAN** algorithm, that is a clustering method that finds groups of similar data points based on how close they are to each other. It works well for accelerometer and gyroscope data because it doesn't expect the data to follow a normal shape. Instead, it looks for dense areas in the data and marks anything far from these areas as an outlier. This makes it great for detecting outliers in sensor data, which often varies a lot.

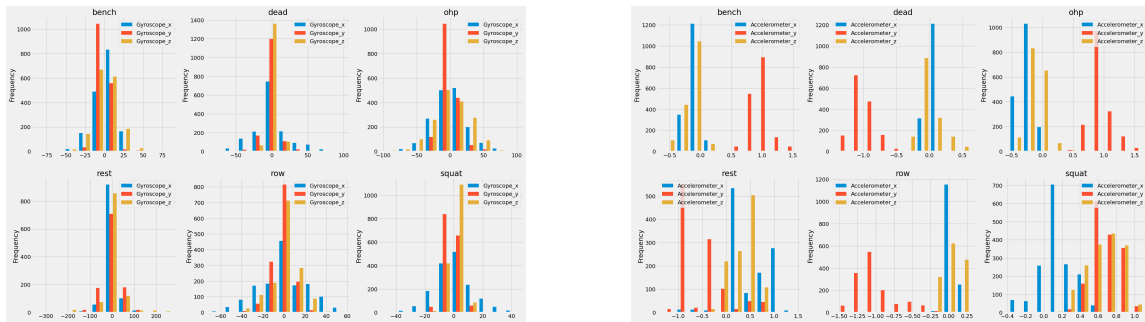


Figure 3: Normal Distribution in Gyroscope, Accelerometer.

As shown in the figure, our sensor data doesn't follow a normal (bell-shaped) distribution.

Dealing with Outliers

After detecting outliers using **DBSCAN**, we do not want to remove them to avoid introducing gaps in the time-series data. Instead, we used **linear interpolation** to estimate and replace the outlier values.

The process worked as follows:

- DBSCAN identified outliers and labeled them with -1.
- For each selected feature (e.g., accelerometer and gyroscope data), we retained the original values and applied interpolation only where outliers were detected.
- **Linear interpolation** estimates the missing or noisy values by drawing a straight line between the values immediately before and after the outlier, filling in the gap with a value along that line.

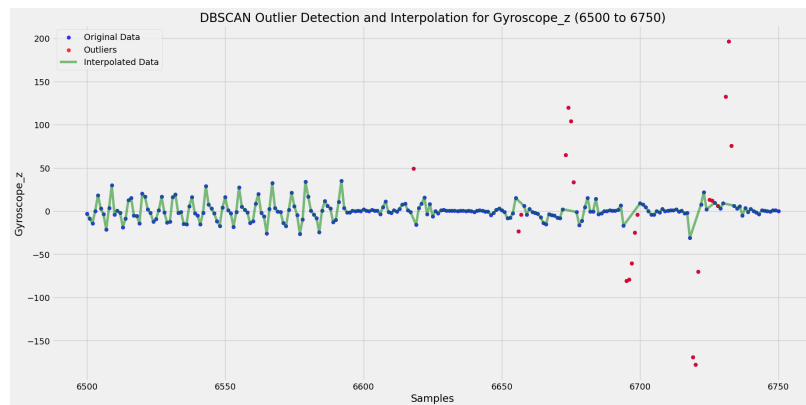


Figure 4: Corrected Signal Using Interpolation After Outlier Removal.

Feature Engineering

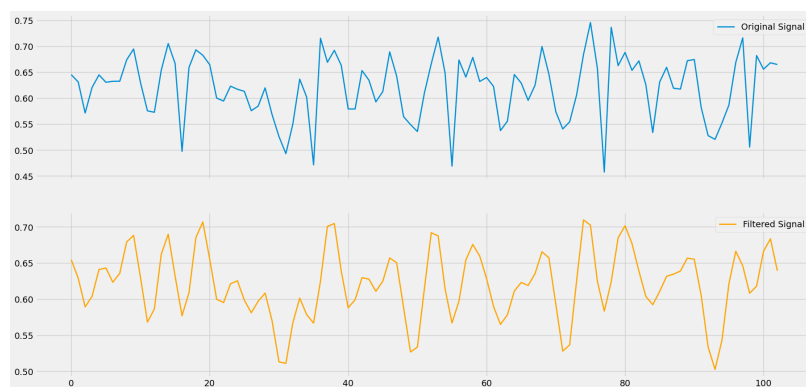


Figure 5: Transformation before and after the filter.

In the feature engineering step, is applied a low-pass filter to smooth the sensor data. The low-pass filter removes unwanted high-frequency noise, like sensor errors and vibrations, while keeping the important slower signals that show real body movement.

The cutoff frequency to 1.2 Hz to keep the slower movements from exercises, like squats, and remove faster noise. By removing the noise, the data becomes easier to analyze, leading to more accurate results.

Feature Extraction

1. Principal Component Analysis (PCA)

Before applying PCA, the data was normalized to ensure a fair comparison between larger and smaller values.

PCA was then applied to reduce the number of features for the gyroscope and accelerometer data. Using the elbow technique, the original six columns were reduced to three principal components: `pca_1`, `pca_2`, and `pca_3`.

These three PCA components were added to the dataset to see how well this technique perform in this case.

2. Magnitude Calculation

To measure the overall strength of motion, we compute the magnitude of the accelerometer and gyroscope signals using the Euclidean norm. This combines the x, y, and z components into a single value:

$$\text{Magnitude} = \sqrt{x^2 + y^2 + z^2}$$

This helps summarize 3D motion into one value for easier analysis, such as classifying different exercises.

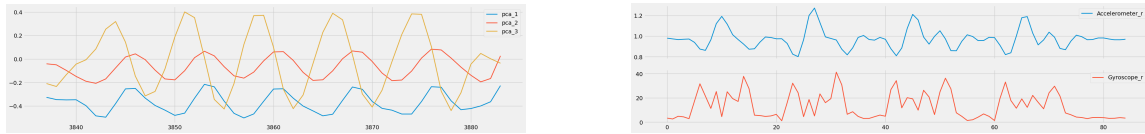


Figure 6: Extracted PCA components and squared magnitudes.

3. Abstract Mean

To make the sensor data smoother and easier to understand, a **moving average** is used. It takes a few values next to each other and finds their average, making the data smoother and less jumpy.

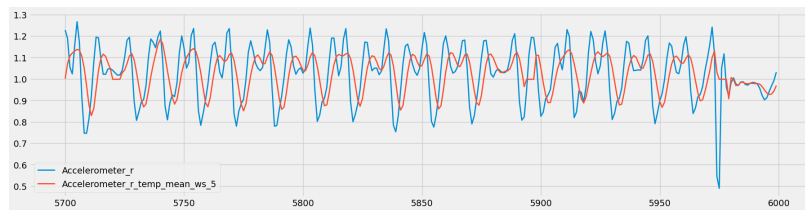


Figure 7: Extracting the abstract mean.

A **window size of 5** is used, meaning every 5 values are averaged together. This reduces noise and shows the overall trend in movement.

The moving average is applied to all sensor values, including the **accelerometer** and **gyroscope** magnitudes, for every data set.

NaN (missing) values can appear at the start, end, or even within the signal. At the start and end, there may be too few values to fill the window. Within the signal, NaNs occur if the original data contains missing values that affect the average calculation.

3.1 Handling NaN Values

To fix this, NaNs are replaced by the mean of each column, calculated while ignoring NaNs. This keeps the data smooth and complete without gaps.

Clustering

k -Means clustering is applied to both accelerometer and gyroscope data to identify patterns in motion signals. The optimal number of clusters is selected using the Elbow Method, helping to group similar movement behaviors.

In Figure 8, each color represents a different cluster, showing how motion data points are separated based on their characteristics. These clusters are then used as labels for training the model to recognize specific types of exercises.

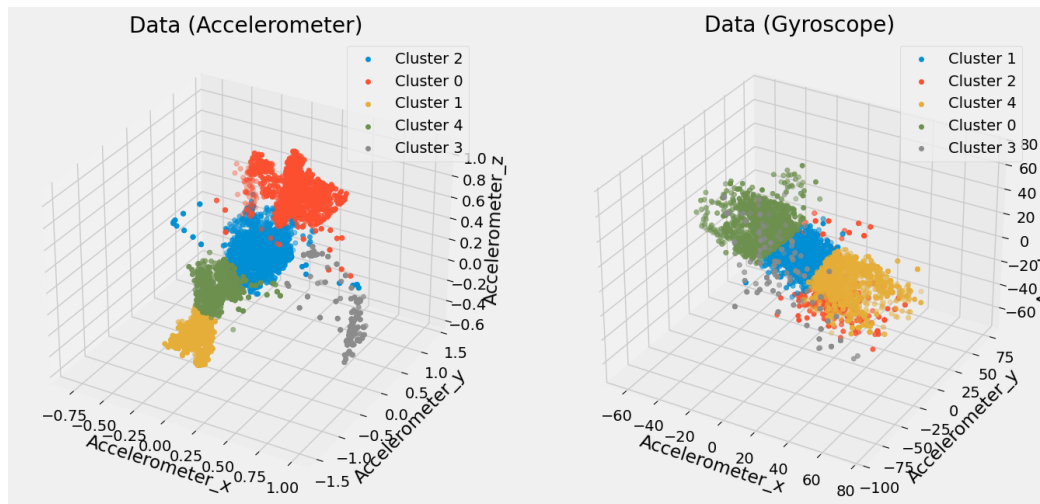


Figure 8: Clustering of accelerometer and gyroscope data.

Feature Correlation

Feature correlation was analyzed to identify variables that are strongly related to each other and to the target behavior.

The heatmap in Figure 9 shows the Pearson correlation coefficients between all extracted features. High correlation values (close to 1 or -1) indicate a strong linear relationship between two features, while values near 0 suggest weak or no correlation.

Two clear examples can be observed:

- `Accelerometer_x` and `Accelerometer_x_temp_mean_ws_5` have a very strong positive correlation ($r = 0.95$), since the temporal mean is directly computed from the original signal.
- `pca_1` shows high correlation with several original sensor features, indicating that the first principal component effectively captures a large portion of the data's variance.

Additionally, the clustering labels (`cluster_acc`, `cluster_gyro`) show moderate correlations with several features, confirming that the clustering process captures meaningful patterns in the data.

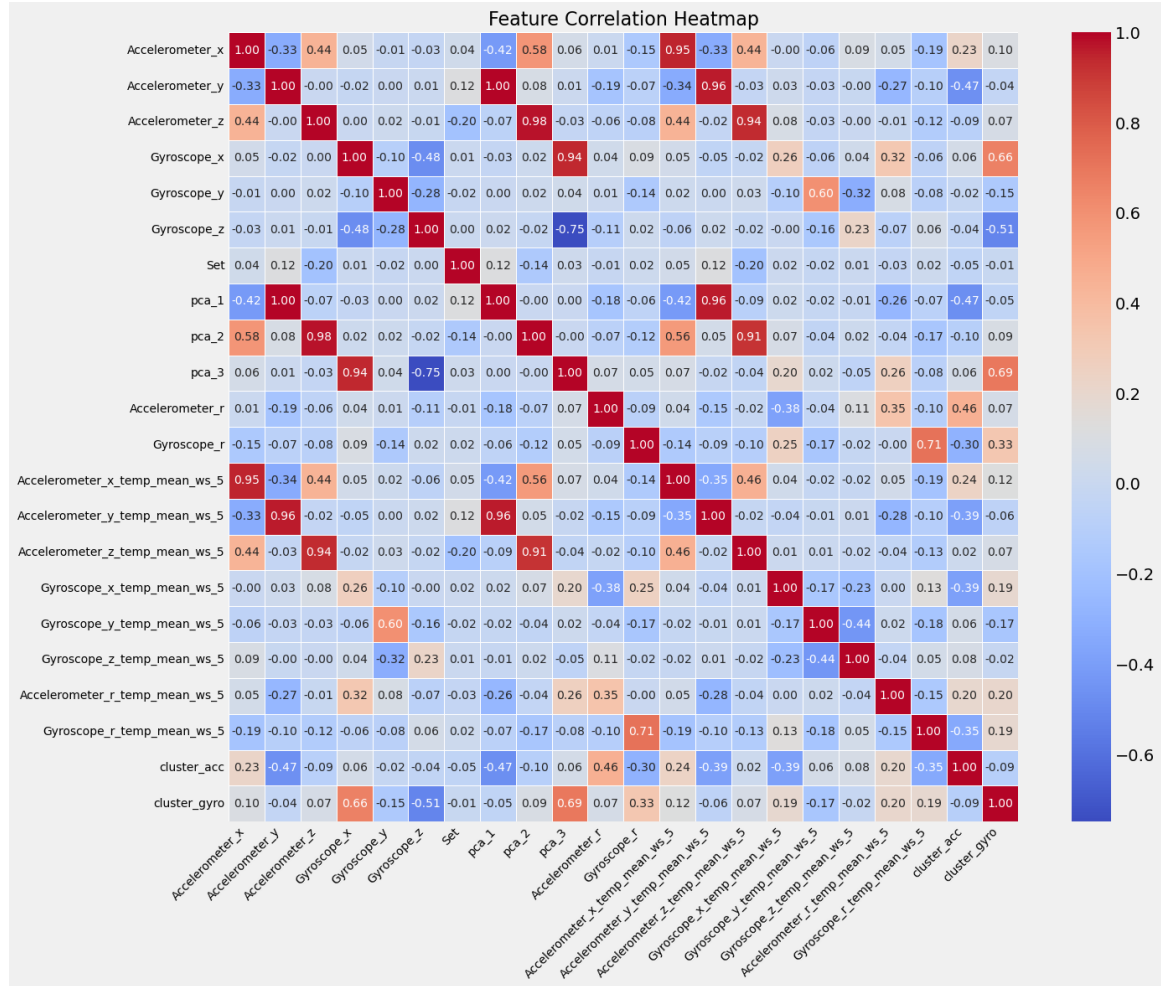


Figure 9: Pearson correlation heatmap.

Modeling Section

The number of samples for each activity was counted to understand how the data is distributed before training the model. The bar chart below shows how often each activity appears in the full dataset (*Total*), and how they are split between the training and test sets.

The data was split using `train_test_split` with `stratify=y` to keep the same activity distribution in both sets. This helps the model learn from all activity types and gives a fair test.

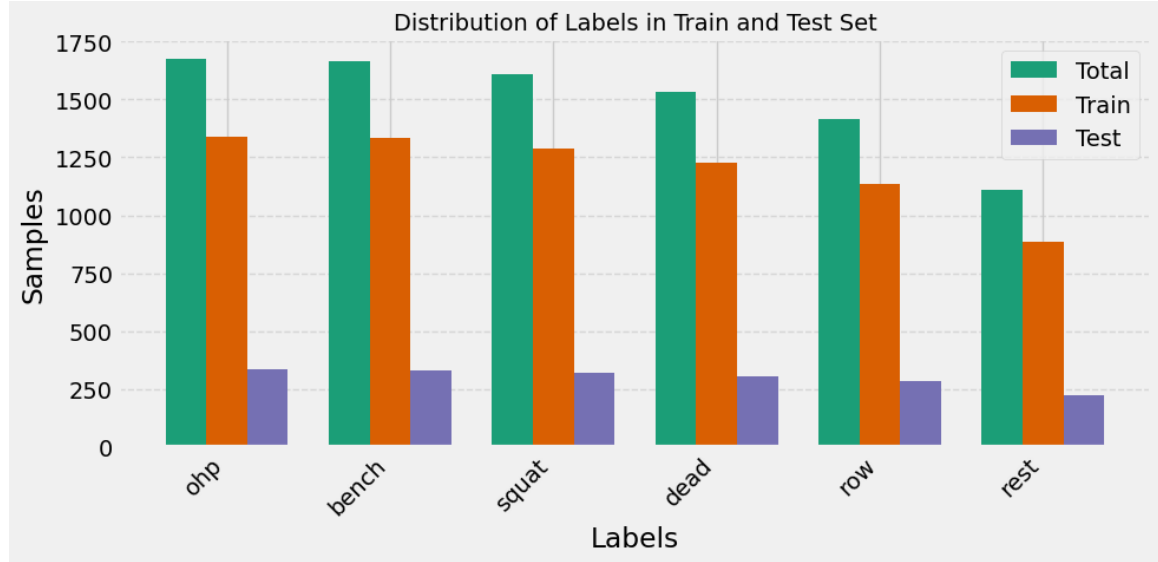


Figure 10: Distribution of activities in training and test sets.

Set Division

The features were grouped into five different sets:

- **Basic Features:** Raw accelerometer and gyroscope signals (x, y, z).
- **Square Features:** Signal magnitudes calculated from the raw sensor data.
- **PCA Features:** Features reduced using Principal Component Analysis (PCA).
- **Cluster Features:** Cluster labels from accelerometer and gyroscope data.
- **PCA-Only Set:** Only PCA features are used instead of accelerometer and gyroscope signals that are removed.

Mean features were initially excluded to observe how the model performs without them. Then, forward feature selection was used to identify the 10 most relevant features, which may include mean-based features.

These five sets are used to compare model performance across different feature combinations and understand the contribution of each type to classification accuracy.

Forward Selection using Decision Tree

A new set of features was selected using forward selection with a decision tree. This method adds one feature at a time, choosing the one that improves model accuracy the most at each step.

- Starts with no features.
- Each remaining feature is tested, and the one that helps the model the most is selected.
- The process repeats until the desired number of features is selected.
- The result is a list of features that improve performance step by step.

This helps identify the most important features for classification.

The final set of selected features is:

```
['pca_1', 'Accelerometer_z', 'Accelerometer_x', 'Gyroscope_r_temp_mean_ws_5',
'Accelerometer_y_temp_mean_ws_5', 'Gyroscope_r', 'Gyroscope_y',
'Accelerometer_x_temp_mean_ws_5', 'Accelerometer_r_temp_mean_ws_5', 'Gyroscope_z']
```

This list includes both original sensor features and mean-based features, showing that mean values can help improve model accuracy.

Results and Discussion

Feature Set Performance:

The best results came from the *selected features* set. It included **mean values**, which helped reduce noise and summarize motion, leading to the highest accuracy (RF: **0.9839**).

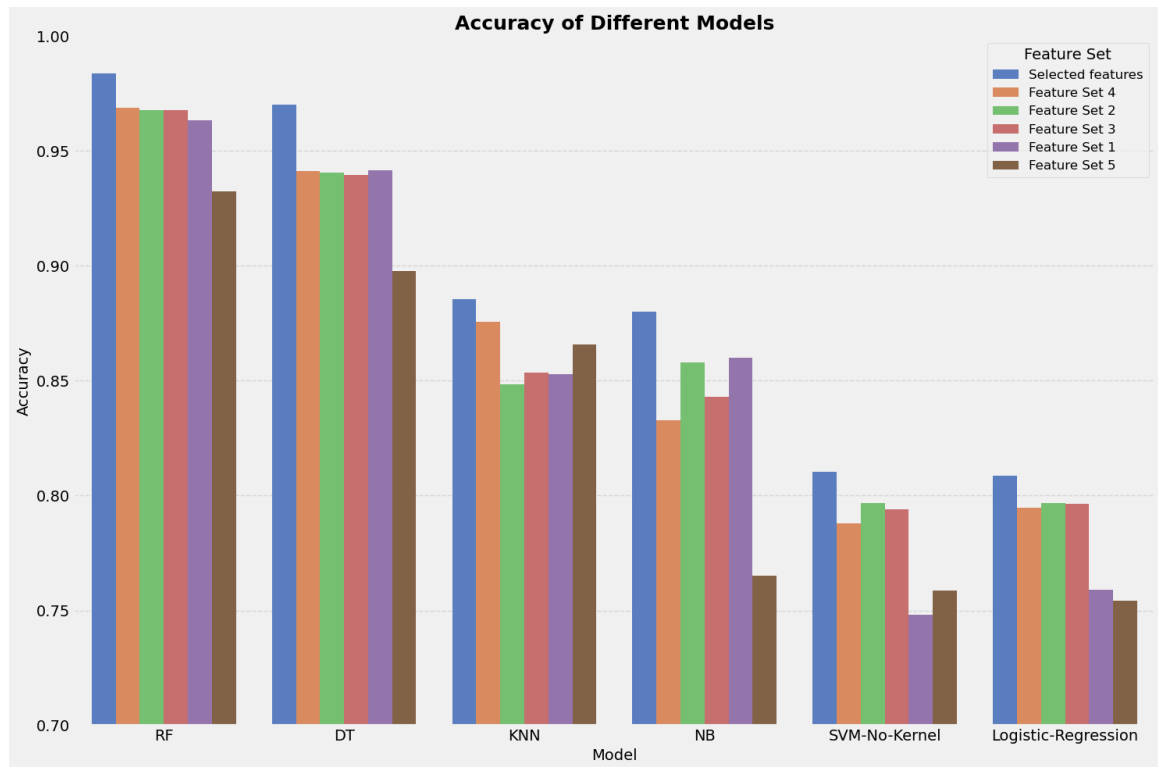


Figure 11: Model performance with different feature sets.

Feature Set 1 used only raw data. It was simple but lacked useful summaries, resulting in lower accuracy (RF: **0.9634**).

Feature Set 2 added squared values, improving signal description slightly.

Feature Set 3 applied PCA on raw data, reducing noise but not adding new information.

Feature Set 4 included cluster features, which added complexity but didn't improve results much.

Feature Set 5 removed raw data and used only PCA. This reduced performance due to missing key features.

Models Performance & Algorithms

DBSCAN: Outlier Detection and Parameter Tuning

DBSCAN was used to detect outliers in the dataset. To improve its performance, the key parameters were carefully studied.

Main Parameters of DBSCAN:

- **eps** (epsilon): The maximum distance between two points to be considered neighbors.
- **min_samples**: The minimum number of neighboring points required to form a dense region (a cluster).

Silhouette Score: This score measures the quality of clustering, ranging from -1 to 1.

A score closer to 1 means the points are well grouped; a score near -1 suggests poor clustering.

Parameter Tuning Process:

- The dataset is first standardized so that all features have similar scales.
- DBSCAN is applied with different combinations of **eps** (0.1 to 2.0) and **min_samples** (5 to 20).
- For each combination, the silhouette score is calculated to assess clustering quality.
- A good combination gives a high silhouette score and less than 20% of the data marked as outliers. This ensures effective clustering while minimizing data loss.

Best Parameters Found:

$$\text{eps} = 1.6, \quad \text{min_samples} = 7$$

This combination achieved the highest silhouette score of 0.4262, indicating well-formed clusters with minimal outliers.

Random Forest (RF)

Random Forest creates many decision trees and combines their results. This helps make better predictions and avoid overfitting.

Tuned parameters:

- **n_estimators**: Number of trees. More trees usually give better results but take longer. Best: 50.

- **min_samples_leaf:** Minimum samples in a leaf. Larger values make the model simpler. Best: 2.

We used `GridSearchCV` to find these best values.

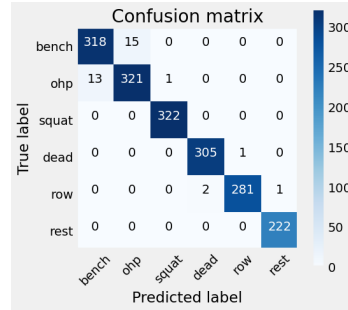


Figure 12: Confusion matrix for Random Forest.

Random Forest performs better than other models on accelerometer and gyroscope data because it builds many trees that each capture different parts of the complex and noisy signals. By combining these trees, the model reduces errors caused by noise and captures subtle patterns, making it especially effective for this kind of sensor data.

Decision Tree (DT)

Decision Trees split data by checking feature values to create a simple tree model.

Key parameter tuned:

- **min_samples_leaf:** Minimum samples needed at a leaf node. Best value: 2.

We used `GridSearchCV` to find the best setting.

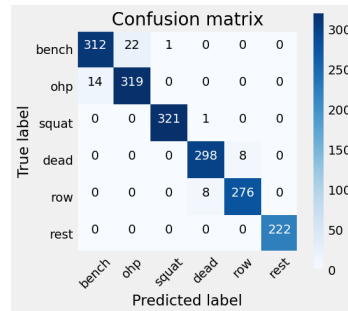


Figure 13: Confusion matrix for Decision Tree.

Decision Tree worked well but not as good as Random Forest. This is because one tree can focus too much on small details or noise in the data. Random Forest uses many trees and combines their results, which helps it handle noise better and make more accurate predictions for sensor data.

K-Nearest Neighbor (KNN)

KNN classifies data points by looking at the majority class of their closest neighbors.

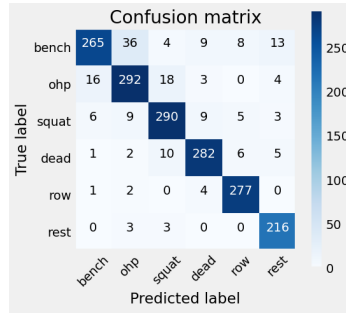


Figure 14: Confusion matrix for KNN.

Key parameter tuned:

- **n_neighbors:** Number of neighbors to consider. Best value: 1.

GridSearchCV was used to find the best value.

Although KNN performed well, it did not classify the data as accurately as Random Forest or Decision Tree. This is because KNN only looks at nearby points and can be easily affected by noisy or overlapping sensor data.

Naive Bayes (NB)

Naive Bayes is a simple model that assumes all features are independent. We did not tune any parameters because it usually works well by default.

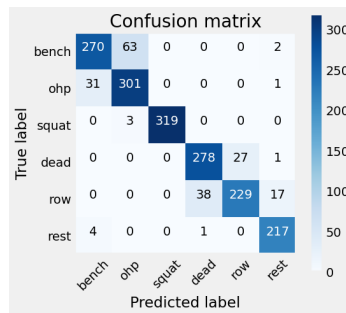


Figure 15: Confusion matrix for NB.

Naive Bayes performed a bit worse than KNN because its assumption that features are independent is not true for sensor data. The features are related, so Naive Bayes cannot capture this well. KNN looks at nearby points, so it handles these relationships better.

Support Vector Machine (SVM) without Kernel

Linear SVM finds a straight line to separate classes. It performed well because it balances fitting the data and keeping the decision boundary simple, helping avoid overfitting on sensor data.

However, it performs worse than tree-based models because the data is complex and not always linearly separable. This limits SVM's ability to capture all the patterns in accelerometer and gyroscope signals.

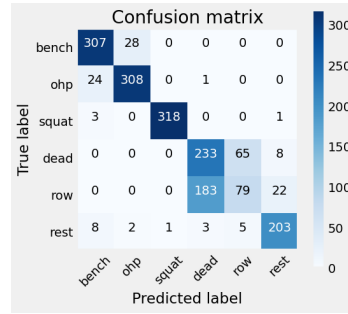


Figure 16: Confusion matrix for SVM No kernel.

Parameters tuned:

- **C**: Controls the trade-off between margin size and classification errors.
- **max_iter**: Number of optimization steps.

GridSearchCV found the best values: **C** = 100, **max_iter** = 1000.

Logistic Regression

Logistic Regression is a simple linear model that predicts class probabilities. It worked well because it models the relationship between features and classes without overfitting, making it suitable for this sensor data.

However, it performs worse than more complex models because it assumes a linear relationship, which is too simple for the noisy and complex patterns in accelerometer and gyroscope data.

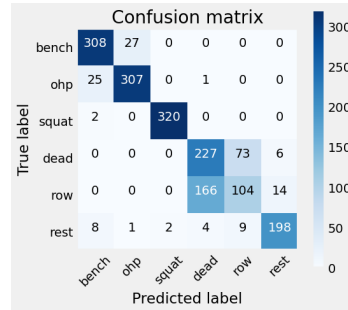


Figure 17: Confusion matrix for LR.

The regularization parameter **C** was tuned with GridSearchCV. Smaller **C** means stronger regularization. Best value: **C** = 1.

Models Comparison

The table below shows how each model performed on different exercises using accelerometer and gyroscope data. The results are stable across precision, recall, and F1-score, indicating that the models are not overfitting.

As we can see, the **Random Forest** model achieved the highest precision among all models. This means it was the most accurate at correctly classifying exercises, making fewer false positives.

In contrast, the **SVM without kernel** also performed well but with lower precision compared to Random Forest. This suggests that while SVM works, it has trouble telling apart exercises with similar movements, like *row* and *deadlift*. These exercises produce similar sensor signals, and both SVM and **Naive Bayes** have difficulty separating them because they can't fully understand complex relationships between features.

Model Metrics by Exercise																			
Model	Class	F1-Score					Precision					Recall					bench	ohp	squat
		bench	ohp	squat	dead	row	rest	bench	ohp	squat	dead	row	rest	bench	ohp	squat	dead	row	rest
DT		0.954	0.951	0.997	0.969	0.968	1.000	0.939	0.969	0.997	0.956	0.982	1.000	0.970	0.934	0.997	0.984	0.954	1.000
KNN		0.848	0.849	0.891	0.884	0.910	0.958	0.828	0.886	0.909	0.883	0.895	0.932	0.868	0.815	0.873	0.886	0.926	0.986
Logistic-Regression		0.912	0.896	0.992	0.640	0.431	0.916	0.910	0.882	0.994	0.557	0.573	0.926	0.913	0.910	0.991	0.752	0.345	0.905
NB		0.873	0.869	0.983	0.831	0.761	0.967	0.849	0.888	0.997	0.772	0.865	0.937	0.898	0.851	0.969	0.899	0.680	1.000
RF		0.961	0.960	0.998	0.990	0.989	1.000	0.958	0.964	0.997	0.993	0.986	1.000	0.964	0.955	1.000	0.987	0.993	1.000
SVM-No-Kernel		0.911	0.900	0.997	0.638	0.388	0.927	0.913	0.887	1.000	0.548	0.564	0.913	0.910	0.913	0.994	0.765	0.296	0.941

Figure 18: Scores.

Conclusion

This project focused on analyzing raw accelerometer and gyroscope data by extracting important features like mean and magnitude and applying filters to clean the signals. Several machine learning models were tested to classify different gym exercises. Among them, Random Forest performed best due to its ability to handle complex and noisy data. The project showed that careful data preparation and feature extraction play a crucial role in improving model accuracy. Overall, the work demonstrates the importance of each step in the machine learning process when working with sensor data, from understanding the signals to selecting and tuning models.

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